

SUMBURGH, United Kingdom

WMO: 030030

Lat: 59.879N

Lon: 1.296W

Elev: 20

StdP: 14.69

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	29.8	31.8	21.2	15.9	33.4	23.2	17.5	34.8	45.3	39.9	42.9	41.1	11.9	350	0.819

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.6	61.7	58.9	60.4	57.7	59.0	56.4	59.4	61.4	58.2	59.9	57.2	58.7	12.2	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.7	73.8	60.8	57.3	70.1	59.0	55.8	66.4	57.6	26.0	61.1	25.0	60.0	24.5	58.3	64.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 40.7	35.3	31.4	26.0	64.4	3.1	2.5	23.8	66.2	22.0	67.6	20.3	69.0	18.1	70.8		
(5)			24.4	61.4	2.6	2.1	22.5	62.9	21.0	64.1	19.5	65.2	17.6	66.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.2	41.3	40.6	41.4	43.8	46.9	51.2	54.8	55.3	53.0	49.6	44.8	42.7
	DBStd	6.36	3.87	3.98	4.02	3.52	3.31	2.51	2.74	2.31	3.20	3.63	4.59	4.83
	HDD50	1554	271	263	268	186	103	15	1	0	12	49	158	228
	HDD65	6516	736	683	733	636	561	414	317	302	360	476	605	693
	CDD50	513	0	0	0	0	7	51	149	163	102	38	3	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	15.3	20.4	18.4	17.8	14.7	12.5	11.6	10.5	11.2	13.9	16.2	17.6	18.4
Precipitation	PrecAvg	45.5	5.7	4.4	4.1	2.4	2.4	2.2	2.4	3.3	3.7	4.7	5.0	5.3
	PrecMax	54.6	10.7	7.2	7.3	4.4	5.1	4.6	4.9	7.5	5.2	6.8	7.3	9.4
	PrecMin	37.2	1.9	1.8	1.1	1.1	0.6	0.5	0.9	1.5	1.3	2.2	2.1	2.4
	PrecStd	4.8	2.0	1.4	1.3	0.7	0.9	1.0	1.0	1.3	1.1	1.3	1.3	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.9	48.6	49.3	52.2	57.4	61.2	64.8	63.0	60.8	56.8	53.8	52.0
		MCWB	48.3	47.0	47.2	48.4	53.7	56.9	61.3	59.6	57.6	54.0	51.5	50.0
	2%	DB	48.4	48.3	48.4	50.4	55.0	57.4	62.5	61.2	59.2	55.6	52.2	50.4
		MCWB	47.2	46.9	46.8	47.5	51.4	54.3	60.1	58.3	56.4	53.2	50.2	48.9
	5%	DB	48.0	46.8	47.8	49.8	53.4	55.8	60.6	60.0	57.6	55.0	51.8	50.0
		MCWB	46.7	45.5	46.3	47.2	50.2	52.8	58.2	57.3	55.3	52.8	49.9	48.6
	10%	DB	46.6	46.3	46.5	48.4	51.9	55.3	59.0	59.0	57.2	53.9	50.4	48.6
		MCWB	45.2	45.0	45.0	46.5	48.8	52.3	56.6	56.4	55.0	51.8	48.5	47.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.6	48.0	47.7	49.8	54.9	58.2	61.9	60.3	58.5	55.1	52.6	50.8
		MCDB	48.6	48.3	48.0	50.6	56.9	61.3	64.2	62.1	60.2	55.9	53.3	51.7
	2%	WB	47.4	46.7	47.0	48.5	52.1	55.1	60.1	59.1	57.3	54.2	51.6	49.5
		MCDB	47.8	47.4	47.5	49.4	54.1	57.2	61.9	61.0	58.9	55.3	52.3	50.1
	5%	WB	46.3	45.8	46.0	47.5	50.7	54.0	58.7	58.0	56.4	53.4	50.4	48.6
		MCDB	47.2	46.7	46.7	48.6	52.5	56.0	60.3	59.8	57.9	54.6	51.3	49.3
	10%	WB	45.3	44.6	45.1	46.5	49.5	53.0	57.2	57.2	55.5	52.7	49.3	47.4
		MCDB	46.4	45.7	46.0	47.9	51.1	54.9	58.7	58.7	57.1	54.1	50.3	48.5
Mean Daily Temperature Range	5% DB	MDBR	5.5	5.4	6.0	6.7	7.5	6.6	6.5	6.6	5.9	5.6	5.7	5.8
		MCDBR	5.9	5.1	5.5	6.6	8.3	7.4	7.8	7.7	6.3	5.1	5.2	5.3
		MCWBR	6.6	5.5	5.3	5.0	5.9	5.3	5.6	5.7	5.0	5.0	6.3	6.3
	5% WB	MCDBR	5.6	4.8	5.0	5.7	7.9	6.9	7.4	7.0	5.7	4.8	5.1	5.2
		MCWBR	6.5	5.6	5.4	5.1	6.0	5.1	5.7	5.5	4.9	5.4	6.5	6.6
Clear-Sky Solar Irradiance	taub	0.338	0.356	0.370	0.387	0.375	0.371	0.385	0.371	0.366	0.351	0.334	0.429	
	taud	2.093	2.208	2.278	2.291	2.394	2.447	2.422	2.461	2.435	2.322	2.151	2.460	
	Ebn at Noon	125	197	238	257	270	273	266	261	240	199	127	87	
	Edh at Noon	16	23	30	35	34	32	32	29	25	21	14	11	
All-Sky Solar Radiation	RadAvg	80	253	615	1117	1561	1580	1444	1177	721	360	116	50	
	RadStd	8	30	41	80	99	151	125	108	60	34	10	4	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (19 neighbors)	+0.58	N/A	+1.40	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page